

Project Name: Technical Analysis for FAANG Stocks

Objective

In this project you will need to implement MA-RSI technical analysis for AAPL, AMZN, NFLX, META and GOOG for the period of the last 10 years. By adjusting the buy-sell point based on technical indicators to improve strategy performance.

Key Steps

- Fetch daily data for these 5 stocks and SP500 from 2014 - 2024.
- Implement strategy of MA-RSI and back test with the dataset, generate annualized return, sharpe ratio and max drawdown.
- Implement different settings of MA-RSI, calculate MA 50 days and 14 days RSI, then calculate short term MA of 10 days and RSI 9 or 25 days to compare results.
- Plot all returns and mark drawdown point, with a base line of SP500 returns.

Bonus

- Improve the MA-RSI indicators with MACD and Bollinger band
- Compare FAANG with other sectors such as Financial or Energy sectors.

Tools and Libraries:

- Language: Python 3.7+
- Data Analysis lib: pandas, numpy
- Data Visualization: matplotlib, seaborn
- Financial Data API: yfinance (to fetch historical data of QQQ) or Stooq

Portfolio Performance Measurement:

- Annualized Return, Sharpe Ratio and Max Drawdown
- Plot the beta return (SPY) over the last 10 years as the baseline
- Plot the portfolio total value as another line to compare with baseline

References

The MA-RSI is a technical analysis strategy that combines the Moving Average (MA) with the Relative Strength Index (RSI) to create a more comprehensive trading signal. Both indicators are widely used in technical analysis for identifying trends, potential reversal points, and overbought or oversold conditions in the market. By combining these two, traders aim to filter out noise and improve the accuracy of their trading signals. Here's a breakdown of each component:

Moving Average (MA)

The Moving Average smooths out price data to create a single flowing line, making it easier to identify the direction of the trend. Moving averages can be simple (SMA) or exponential (EMA), among other types, with the primary difference being the weighting of the data points. Traders often use two MAs with different time periods (a shorter and a longer one) to identify potential trend reversals, signaled by the crossover of these lines.

Relative Strength Index (RSI)

The RSI is a momentum oscillator that measures the speed and change of price movements on a scale of 0 to 100. It is typically used to identify overbought or oversold conditions. An asset is generally considered overbought when the RSI is above 70 and oversold when it's below 30. The RSI can also show divergence signals, indicating potential reversals when the indicator moves in the opposite direction of price.

Combining MA with RSI (MA-RSI Strategy)

The MA-RSI strategy leverages the strengths of both indicators to provide more reliable signals for entering or exiting trades. Here's how:

1. **Trend Identification with MA:** Use the moving average (or moving averages in the case of a dual MA system) to identify the overall trend direction. A common approach is to consider the market in an uptrend when the price is above its moving average and in a downtrend when below.
Overbought/Oversold Conditions with RSI: Use the RSI to pinpoint potential entry or exit points based on overbought or oversold conditions. For instance, in an uptrend, look for the RSI to fall below 30 (oversold) and then move back above this level as a potential buy signal. Conversely, in a downtrend, an RSI above 70 (overbought) dropping back below might be considered a sell signal.
Confirmation and Filtering: The MA helps to confirm the trend direction, while the

RSI provides specific entry or exit points. This combination helps to filter out less probable signals by focusing on those that align with both the trend (as shown by the MA) and momentum (as indicated by the RSI).

Application

- **Buy Signal:** During an uptrend (price above MA), a buy signal might be generated when the RSI moves from oversold (below 30) back above 30, indicating potential upward momentum resuming.
Sell Signal: During a downtrend (price below MA), a sell signal might be indicated when the RSI moves from overbought (above 70) back below 70, suggesting downward momentum resuming.

Limitations

While the MA-RSI strategy can be powerful, it's important to remember that no indicator or strategy guarantees success. False signals can occur, and market conditions can change rapidly. It's beneficial to use this strategy in conjunction with other analysis tools, risk management techniques, and a clear trading plan.

MACD

The Moving Average Convergence Divergence (MACD) is a trend-following momentum indicator that shows the relationship between two moving averages of a security's price. The MACD is calculated by subtracting the 26-period Exponential Moving Average (EMA) from the 12-period EMA. The result of that calculation is the MACD line. A nine-day EMA of the MACD called the "signal line," is then plotted on top of the MACD line, which can function as a trigger for buy and sell signals.

Components of MACD

- **MACD Line:** 12-period EMA minus 26-period EMA.
Signal Line: 9-period EMA of the MACD Line.
Histogram: The difference between the MACD Line and the Signal Line.

How MACD is Used

- Trend Identification: The MACD line's crossover above or below the signal line can indicate bullish or bearish trends.

Momentum and Reversals: The divergence between the MACD and the price suggests potential reversals. Also, the histogram provides a visual representation of the momentum (the wider the gap, the stronger the momentum).